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Pump valve module and method for operating the same, solenoid valve-based pump valve module for providing massage function for vehicle seat, and vehicle seat

Abstract

A pump valve module and operating method, a solenoid valve-based pump valve module for providing a massage function for a vehicle seat, and a vehicle seat are disclosed. A composite solenoid valve in the pump valve module includes deflation solenoid valve and inflation solenoid valve connected in series and formed integrally by injection molding. A first part region of the inflation solenoid valve is connected to air source. A second part region of the deflation solenoid valve is connected to inflated bag. A second part region of the inflation solenoid valve is connected to a first part region of the deflation solenoid valve. Inflation is implemented when the inflation solenoid valve is powered on and the deflation solenoid valve is de-energized. Deflation is implemented through a third part region of the deflation solenoid valve when the inflation solenoid valve is de-energized and the deflation solenoid valve is powered on.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present disclosure claims priority to Chinese Patent Application No. 201911107091.8, filed with the Chinese Patent Office on Nov. 13, 2019, entitled “Composite Solenoid Valve, Pump Valve Module, and Vehicle Seat”, Chinese Patent Application No. 202010364943.8, filed with the Chinese Patent Office on Apr. 30, 2020, entitled “Solenoid Valve-based Pump Valve Module for Providing Massage Function for Vehicle Seat, and Vehicle Seat”, and Chinese Patent Application No. 202020707199.2, filed with the Chinese Patent Office on Apr. 30, 2020, entitled “Solenoid Valve-based Pump Valve Module for Providing Massage Function for Vehicle Seat, and Vehicle Seat”, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(2) The present disclosure relates to the field of composite solenoid valves, and in particular to a pump valve module and a method for operating the same, a solenoid valve-based pump valve module for providing a massage function for a vehicle seat, and a vehicle seat.

BACKGROUND ART

(3) In the prior art systems such as pneumatic support, pneumatic massage, and pneumatic adjustment systems for a seat, when multiple inflatable bags are simultaneously connected to one air source, these inflatable bags should be deflated and inflated simultaneously. Therefore, in the use of the seat, one air source is often shared by one or two inflatable bags with the same function. In the case of multiple inflatable bags, a solenoid valve cannot individually control the inflation, deflation, and pressure maintenance of a single inflatable bag, but can only control the inflatable bags simultaneously. Its operation is not flexible and cannot achieve the effects required by humans. If the multiple inflatable bags are individually equipped with and controlled by corresponding pumps and solenoid valves, high cost is required, and a large space is occupied, thus such arrangement is not suitable for small space.

SUMMARY

(4) In view of the above-mentioned disadvantages or drawbacks in the prior art, the present disclosure provides a composite solenoid valve-based pump valve module for a vehicle seat, so as to achieve at least one of the technical effects of differently controlling multiple inflatable bags with one air source.

(5) The present disclosure provides a pump valve module, comprising an air pump and a composite solenoid valve. Each air outlet of the air pump is equipped with one composite solenoid valve.

(6) The composite solenoid valve includes a deflation solenoid valve and an inflation solenoid valve connected in series and formed integrally by injection molding. A first part region of the inflation solenoid valve is connected to an air outlet of an air source. A second part region of the deflation solenoid valve is connected to an inflated bag. A second part region of the inflation solenoid valve is connected to a first part region of the deflation solenoid valve. Inflation is implemented when the inflation solenoid valve is powered on and the deflation solenoid valve is de-energized. The deflation is implemented through a third part region of the deflation solenoid valve when the inflation solenoid valve is de-energized and the deflation solenoid valve is powered on.

(7) The first part region of the inflation solenoid valve communicates with the air outlet of the air pump. At least one composite solenoid valve is provided and is integrated and mounted in a housing formed integrally with the air pump.

(8) Optionally, the second part region of the inflation solenoid valve is located at a structural connecting portion in the middle of the composite solenoid valve and is connected to and staggered axially with respect to the first part region of the deflation solenoid valve, and they are connected

by an injection-molded core. A three-way port is provided in the injection-molded core. The three-way port has an end connected to the second part region of the inflation solenoid valve, an end connected to the first part region of the deflation solenoid valve, and the other end sealed by a sealing ball.

(9) Optionally, the inflation solenoid valve is a two-position two-way solenoid valve or a two-position three-way solenoid valve, and the deflation solenoid valve is a two-position three-way solenoid valve.

(10) Optionally, an air inlet nozzle is mounted in the first part region of the inflation solenoid valve, and the air inlet nozzle is connected to a side wall of the injection-molded core via a U-shaped iron.

(11) Optionally, an air outlet nozzle is mounted in the third part region of the deflation solenoid valve, and a side wall of the air outlet nozzle is connected to a side wall of the injection-molded core via a U-shaped iron.

(12) Optionally, a metal valve core of the inflation solenoid valve and/or the deflation solenoid valve is of one of a flat type and a spline type.

(13) Optionally, the pump valve module further includes a pressure sensor. The pressure sensor is mounted in an air path of the pump valve module or mounted inside the composite solenoid valve.

(14) The present disclosure provides an operating method based on a pump valve module, comprising the following working states: a first working condition, where the inflation solenoid valve is powered on, the deflation solenoid valve is de-energized, the inflation solenoid valve is energized, and a metal valve core is moved down, so that a gas enters the inflation solenoid valve from the air source through a connecting channel, enters the first part region of the deflation solenoid valve from the second part region of the inflation solenoid valve through a channel in an injection-molded core, and enters the inflated bag from the second part region of the deflation solenoid valve; a second working condition, where the inflation solenoid valve is de-energized, the deflation solenoid valve is powered on, the deflation solenoid valve is energized, a metal valve core is moved upward, and the connecting channel is closed, so that the gas enters the deflation solenoid valve from the inflated bag and is discharged from the third part region of the deflation solenoid valve through the second part region of the deflation solenoid valve; a third working condition, where the inflation solenoid valve is de-energized, the deflation solenoid valve is de-energized, and the deflation solenoid valve is closed and configured to maintain air pressure in the inflated bag.

(15) The present disclosure provides a vehicle seat, comprising a lumbar support and/or side wing supporting structure having an air source supplied by a pump valve module.

(16) In summary, the present disclosure includes at least the following advantageous effects:

(17) One or more such composite solenoid valves may be arranged in parallel and mounted on the upper part of an air pump to form a pump valve module together with the air pump. When the inflation solenoid valve of the composite solenoid valve is powered on and the deflation solenoid valve is de-energized, the inflation solenoid valve is energized and the valve core moves down to achieve inflation. When the inflation solenoid valve is de-energized and the deflation solenoid valve is powered on, the deflation solenoid valve is energized and the valve core moves upward to achieve deflation. Air pressure in the inflatable bag is maintained when the inflation solenoid valve is de-energized and the deflation solenoid valve is de-energized. The above-mentioned pump valve module enables different control of multiple inflatable bags with one air source, and an air pressure maintaining function is added to the inflatable bags. Moreover, the entire air source delivery and control device is configured as an integrated pump valve module, which has a compact structure, can be easily mounted, has stable functions, and enables the cooperation of multiple units with multiple functions.

(18) An embodiment of the present disclosure provides a solenoid valve-based pump valve module for providing a massage function for a vehicle seat, which includes an air distribution layer and further includes a solenoid valve hermetically connected to an air outlet of the air distribution layer. The solenoid valve includes a valve body. A solenoid coil is provided around a side wall of the

valve body. The valve body is provided with an air distribution chamber, and the two ends of the air distribution chamber act as an air inlet and an air discharge port, respectively. The side wall of the valve body is provided with a branched air passage communicating with the air distribution chamber and facing the same side as the air discharge port. The air discharge port is equipped with an air discharging member in which an air passage is provided, and a valve core built in the air distribution chamber is provided under the air discharging member. A first elastic element is sleeved on both the air discharging member and the valve core. The presence or absence of a gap between the air discharging member and the valve core can be adjusted by switching the solenoid coil to an energized or de-energized state.

(19) For the solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to the present disclosure, it is possible to adjust whether the air discharging member and the valve core are attracted to each other by adjusting whether the solenoid coil is in an energized state, and finally it is possible to adjust whether there is a gap between the air discharging member and the valve core by switching the solenoid coil to the energized or de-energized state. When the solenoid coil is switched to the energized state, there is no gap between the air discharging member and the valve core, and there is a gas flow in the inflation passageway. When the solenoid coil is switched to the de-energized state, there is a gap between the air discharging member and the valve core, and there is a gas flow in the deflation passageway. The working state of the inflatable bag can be adaptively switched from the inflated state to the deflated state by switching the state of the inflation passageway to the state of the deflation passageway. The branched air passage is on the same side as the air discharge port, so that the inflation end and the deflation end are located on the same side, whereby the overall structure of the solenoid valve can be reduced. In this way, the size of the solenoid valve-based pump valve module is further reduced, and the technical difficulty in further reducing the size of the solenoid valve-based pump valve module is solved.

(20) Optionally, the air distribution layer is provided with at least three air outlets. At least three solenoid valves are provided and are hermetically connected to the air outlets in one-to-one correspondence.

(21) Optionally, the branched air passage includes: an air introducing passage communicating with the air distribution chamber and an air injection passage communicating with a side wall of the air introducing passage. A sealing member is configured in a free end of the air introducing passage.

(22) The structure of the branched air passage is improved to avoid the influence of the machining technology and facilitate the machining of the air introducing passage. A through hole is provided at an end of the air introducing passage farther away from the solenoid valve to facilitate provision of a gas channel in the air introducing passage. A sealing member is configured in the free end of the air introducing passage, so that the air introducing passage can be easily sealed after the machining is finished.

(23) Optionally, an accommodating cavity is provided annularly in an outer wall of the valve body, and the solenoid coil is wound in the accommodating cavity.

(24) Optionally, a first stop groove and a second stop groove are provided annularly in the outer wall of the air discharging member and in the outer wall of the valve core, respectively. The two ends of the first elastic element are locked in the first stop groove and the second stop groove, respectively.

(25) The two ends of the first elastic element are respectively locked in the first stop groove and the second stop groove by fixing the first elastic element in the stop grooves.

(26) Optionally, the pump valve module further includes a first U-shaped iron longitudinally straddling the outer wall of the valve body. The first U-shaped iron is provided with through holes allowing the air inlet and the free end of the air discharging member to be exposed.

(27) Optionally, at least one air outlet of the air distribution layer is equipped with a composite solenoid valve. The composite solenoid valve includes a deflation solenoid valve and an inflation

solenoid valve connected in series and formed integrally by injection molding. A first part region of the inflation solenoid valve is connected to an air outlet of an air source. A second part region of the deflation solenoid valve is connected to an inflated bag. A second part region of the inflation solenoid valve is connected to a first part region of the deflation solenoid valve. Inflation is implemented when the inflation solenoid valve is powered on and the deflation solenoid valve is de-energized. The deflation is implemented through a third part region of the deflation solenoid valve when the inflation solenoid valve is de-energized and the deflation solenoid valve is powered on.

(28) At least one air outlet of the air distribution layer is equipped with a composite solenoid valve. When the inflation solenoid valve of the composite solenoid valve is powered on and the deflation solenoid valve is de-energized, the inflation solenoid valve is energized and the valve core moves down to achieve inflation. When the inflation solenoid valve is de-energized and the deflation solenoid valve is powered on, the deflation solenoid valve is energized and the valve core moves upward to achieve deflation. Air pressure in the inflatable bag is maintained when the inflation solenoid valve is de-energized and the deflation solenoid valve is de-energized. The above-mentioned composite solenoid valve enables different control of multiple inflatable bags with one air source, and an air pressure maintaining function is added to the inflatable bags.

(29) Optionally, the air distribution layer is further provided with a pressure regulating protrusion, and a pressure regulating groove is provided in the pressure regulating protrusion. A pressure regulating hole communicating with the air distribution layer is provided in the bottom of the pressure regulating groove. An overflow valve is configured in the pressure regulating protrusion.

(30) Optionally, the overflow valve includes: an overflow column with a bottom extending into the pressure regulating hole, a limit ring provided around an outer wall of the overflow column, a second elastic element sleeved on a side wall of the overflow column and located above the limit ring, and a top cover sleeved on the top of the overflow column and provided with a through slot in its center. A second sealing ring is sleeved on the side wall of the overflow column under the limit ring.

(31) In the structure of this module, a pressure regulating protrusion communicating with the inside of the air distribution layer is provided in the middle of the air distribution layer, and an overflow valve with a pressure stabilizing function is configured in the pressure regulating protrusion, in order to ensure a stable gas pressure in the air distribution layer. When the air pressure in the air distribution layer is greater than a safety threshold, the gas overflows from the overflow valve so that the air pressure is restored to the safety threshold or lower.

(32) An embodiment of the present disclosure provides a solenoid valve-based pump valve module for providing a massage function for a vehicle seat, which includes an air distribution layer and further includes a solenoid valve hermetically connected to an air outlet of the air distribution layer. The solenoid valve includes a valve body. A solenoid coil is provided around a side wall of the valve body. The valve body is provided with an air distribution chamber, and the two ends of the air distribution chamber act as an air inlet and an air discharge port, respectively. The side wall of the valve body is provided with a branched air passage communicating with the air distribution chamber and facing the same side as the air discharge port. The air discharge port is equipped with an air discharging member in which an air passage is provided, and a valve core built in the air distribution chamber is provided under the air discharging member. A first elastic element is sleeved on both the air discharging member and the valve core. The presence or absence of a gap between the air discharging member and the valve core can be adjusted by switching the solenoid coil to an energized or de-energized state. At least one air outlet of the air distribution layer is equipped with an air path communicating therewith, and the air path has a sealable free end.

(33) Optionally, the air distribution layer is provided with at least four air outlets. At least three of the air outlets are hermetically connected to the solenoid valves in one-to-one correspondence. One of the air outlets is equipped with an air path communicating therewith, and a free end of the air

path is provided with a sealing cover.
(34) In the specific structure of the above-mentioned module in an embodiment of the present disclosure, an air path connected to at least one air outlet of the air distribution layer and communicating therewith is also designed to be reserved for inflating other components. On the other hand, the design of the structure of the solenoid valve is optimized so that the inflation end and the deflation end are located on the same side, whereby the overall structure of the solenoid valve can be reduced, and hence the size of the solenoid valve-based pump valve module is further reduced.
(35) In this technical solution, it is possible to adjust whether the air discharging member and the valve core are attracted to each other by adjusting whether the solenoid coil is in an energized state, and finally it is possible to adjust whether there is a gap between the air discharging member and the valve core by switching the solenoid coil to the energized or de-energized state. When the solenoid coil is switched to the energized state, there is no gap between the air discharging member and the valve core, and there is a gas flow in the inflation passageway. When the solenoid coil is switched to the de-energized state, there is a gap between the air discharging member and the valve core, and there is a gas flow in the deflation passageway. The working state of the inflatable bag can be adaptively switched from the inflated state to the deflated state by switching the state of the inflation passageway to the state of the deflation passageway. The branched air passage is on the same side as the air discharge port, so that the inflation end and the deflation end are located on the same side.
(36) Based on the above-mentioned basic design, the structure of the solenoid valve is further optimized in this technical solution. For example, the structure of the branched air passage is further optimized, and the influence of the machining technology is taken into consideration in order to facilitate the machining of the air introducing passage. A through hole is provided at an end of the air introducing passage farther away from the solenoid valve to facilitate provision of a gas channel in the air introducing passage. A sealing member is configured in the free end of the air introducing passage, so that the air introducing passage can be easily sealed after the machining is finished. For another example, the two ends of the first elastic element are respectively locked in the first stop groove and the second stop groove, in order to easily fix the first elastic element.
(37) In this technical solution, the structure of the air distribution layer of the solenoid valve-based pump valve module is further optimized. In other words, in the structure of this module, a pressure regulating protrusion communicating with the inside of the air distribution layer is provided in the middle of the air distribution layer, and an overflow valve with a pressure stabilizing function is configured in the pressure regulating protrusion, in order to ensure a stable gas pressure in the air distribution layer. When the air pressure in the air distribution layer is greater than a safety threshold, the gas overflows from the overflow valve so that the air pressure is restored to the safety threshold or lower. In addition, the specific structure of the overflow valve is also disclosed in this embodiment.
(38) In this technical solution, the structure of the air distribution layer of the solenoid valve-based pump valve module is further optimized. At least one air outlet of the air distribution layer is equipped with a composite solenoid valve. Based on this design, in this embodiment, when the inflation solenoid valve of the composite solenoid valve is powered on and the deflation solenoid valve is de-energized, the inflation solenoid valve is energized and the valve core moves down to achieve inflation. When the inflation solenoid valve is de-energized and the deflation solenoid valve is powered on, the deflation solenoid valve is energized and the valve core moves upward to achieve deflation. Air pressure in the inflatable bag is maintained when the inflation solenoid valve is de-energized and the deflation solenoid valve is de-energized. The above-mentioned composite solenoid valve enables different control of multiple inflatable bags with one air source, and an air pressure maintaining function is added to the inflatable bags.
(39) The present disclosure provides a vehicle seat, comprising: a solenoid valve-based pump valve

module for providing a massage function for a vehicle seat.
(40) The solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to this technical solution is applicable to a vehicle seat, so that the vehicle seat has a more compact internal space structure so as to form an optimized structural layout.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Other features, objects, and advantages of the present disclosure will become more apparent by reading the detailed description of non-limiting embodiments with reference to the following drawings:
(2) FIG. **1** is a schematic structural diagram of a composite solenoid valve in a pressure maintaining state in the present disclosure;
(3) FIG. **2** is a schematic structural diagram of a pump valve module in the present disclosure;
(4) FIG. **3** is a schematic structural diagram of a composite solenoid valve in an air feeding state according to an embodiment of the present disclosure;
(5) FIG. **4** is a schematic structural diagram of a composite solenoid valve in an air discharging state according to an embodiment of the present disclosure;
(6) FIG. **5** is a sectional view of a metal valve core of a composite solenoid valve in the present disclosure;
(7) FIG. **6** is a schematic structural diagram of a composite solenoid valve with a connecting channel in the present disclosure;
(8) FIG. **7** is a schematic structural diagram of an embodiment of a solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to the present disclosure;
(9) FIG. **8** is a schematic structural diagram of another embodiment of a solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to the present disclosure;
(10) FIG. **9** is a schematic structural diagram of an embodiment of a solenoid valve in an air feeding state according to the present disclosure;
(11) FIG. **10** is a schematic structural diagram of an embodiment of a solenoid valve in an air discharging state according to the present disclosure;
(12) FIG. **11** is a schematic structural diagram of an embodiment of a solenoid valve of the present disclosure;
(13) FIG. **12** is a schematic structural diagram of an embodiment of a solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to the present disclosure;
(14) FIG. **13** is a schematic structural diagram of an embodiment of a solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to the present disclosure.
(15) FIG. **14** is a schematic structural diagram of another embodiment of a solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to the present disclosure;
(16) FIG. **15** is a schematic structural diagram of an embodiment of a solenoid valve in a de-energized state according to the present disclosure;
(17) FIG. **16** is a schematic structural diagram of an embodiment of a solenoid valve in an energized state according to the present disclosure;
(18) FIG. **17** is a schematic structural diagram of an embodiment of a solenoid valve of the present disclosure;

(19) FIG. **18** is a schematic structural diagram of an embodiment of a solenoid valve of the present disclosure;
(20) FIG. **19** is a schematic structural diagram of an embodiment of a solenoid valve-based pump valve module of the present disclosure;
(21) FIG. **20** is a schematic structural diagram of an embodiment of a composite solenoid valve in a pressure maintaining state according to the present disclosure;
(22) FIG. **21** is a schematic structural diagram of an embodiment of a composite solenoid valve in an air discharging state according to the present disclosure;
(23) FIG. **22** is a schematic structural diagram of an embodiment of a composite solenoid valve in an air feeding state according to the present disclosure;
(24) FIG. **23** is a schematic structural diagram of an embodiment of a solenoid valve-based pump valve module of the present disclosure;
(25) FIG. **24** is a schematic structural diagram of an embodiment of a solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to the present disclosure;
(26) FIG. **25** is a schematic partially enlarged structural diagram of an overflow valve of FIG. **21** in the present disclosure; and
(27) FIG. **26** is a schematic structural diagram of another embodiment of a solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to the present disclosure.
(28) Reference Numerals in the Figures: **4**—air pump; **5**—pressure sensor; **7**—air nozzle; **8**—frame; **11**—U-shaped iron; **12**—sealing ball; **13**—upper shell; **14**—PCB board; **10**—solenoid valve; **101**—valve body; **110**—accommodating cavity; **111**—first U-shaped iron; **102**—solenoid coil; **103**—air distribution chamber; **104**—branched air passage; **1041**—air introducing passage; **1042**—air injection passage; **105**—air passage; **106**—air discharging member; **1061**—first stop groove; **1071**—second stop groove; **107**—valve core; **1072**—sealing groove; **108**—first elastic element; **109**—sealing member; **20**—solenoid valve-based pump valve module for providing a massage function for a vehicle seat; **21**—pressure regulating protrusion; **211**—pressure regulating groove; **212**—pressure regulating hole; **22**—overflow valve; **221**—overflow column; **222**—limit ring; **223**—second elastic element; **224**—top cover; **225**—second sealing ring; **23**—air distribution layer; **30**—inflatable bag; **40**—composite solenoid valve; **401**—deflation solenoid valve; **402**—inflation solenoid valve; **403**—injection-molded core; **406**—metal valve core; **409**—spring; **410**—silicone cap; **411**—second U-shaped iron; **415***a*—first part region of a deflation solenoid valve **401**; **415***b*—first part region of a inflation solenoid valve **402**; **416**—connecting channel; **417***a*—second part region of the deflation solenoid valve **401**; **417***b*—second part region of the inflation solenoid valve **402**; **418**—third part region.
DETAILED DESCRIPTION OF THE EMBODIMENTS
(29) The present disclosure will be described in further detail below with reference to the accompanying drawings and embodiments. It will be understood that specific embodiments described here are only intended to explain a relevant invention, but not intended to limit the invention. In addition, it should be noted that only the parts related to the invention are shown in the drawings for ease of description.
(30) It should be noted that the embodiments in the present disclosure and the features in the embodiments can be combined with each other without conflict. The present disclosure will be described below in detail with reference to the accompanying drawings and in connection with embodiments.
(31) Referring to FIG. **1**, a composite solenoid valve according to this embodiment includes a deflation solenoid valve **401** and an inflation solenoid valve **402** connected in series and formed integrally by injection molding. A first part region **415***b* of the inflation solenoid valve **402** is connected to an air outlet of an air source. A second part region **417***a* of the deflation solenoid

valve **401** is connected to an inflated bag or a bag to be inflated. A second part region **417***b* of the inflation solenoid valve **402** is connected to a first part region **415***a* of the deflation solenoid valve **401**. Inflation is implemented when the inflation solenoid valve **402** is powered on and the deflation solenoid valve **401** is de-energized. The deflation is implemented through a third part region **418** of the deflation solenoid valve **401** when the inflation solenoid valve **402** is de-energized and the deflation solenoid valve **401** is powered on. The composite solenoid valve is placed at the top end of the air source and may constitute one structural body together with the air source. Here, the inflation solenoid valve **402** is a two-position two-way solenoid valve. The second part region **417***b* of the inflation solenoid valve **402** is configured to allow air to be fed into the deflation solenoid valve **401**. The first part region **415***b* of the inflation solenoid valve **402** is configured to allow air to be fed from the air source. The deflation solenoid valve **401** is a two-position three-way solenoid valve. The second part region **417***a* of the deflation solenoid valve **401** is configured to inflate the inflated bag. The third part region **418** of the deflation solenoid valve **401** is configured to release the air. The first part region **415***a* of the solenoid valve **401** is configured to receive the gas from the inflation solenoid valve **402**. The composite solenoid valve is placed at the top end of the air source to form one structural body so as to achieve an integrated compact structural arrangement.
(32) Here, the first part region **415***a* of the deflation solenoid valve **401** acts as the air inlet of the deflation solenoid valve **401**, the second part region **417***a* of the deflation solenoid valve **401** acts as the air outlet of the deflation solenoid valve **401**, and the third part region of the deflation solenoid valve **401** **418** acts as the deflation port of the deflation solenoid valve **401**. The first part region **415***b* of the inflation solenoid valve **402** acts as the air inlet of the inflation solenoid valve **402**. The second part region **417***b* of the inflation solenoid valve **402** acts as the air outlet of the inflation solenoid valve **402**. Here, the type of the air source may be selected from an air pump, an air source carried in the body of a vehicle, or other components that can be used as the air source.
(33) The second part region **417***b* of the inflation solenoid valve **402** is located at a structural connecting portion in the middle of the composite solenoid valve and is connected to and staggered axially with respect to the first part region **415***a* of the deflation solenoid valve **401**, and they are connected by an injection-molded core **403**. A three-way port is provided in the injection-molded core **403**. The three-way port has an end connected to the second part region **417***b* of the inflation solenoid valve **402**, an end connected to the first part region **415***a* of the deflation solenoid valve **401**, and the other end sealed by a sealing ball **12**. Here, the second part region **417***b* of the inflation solenoid valve **402** is staggered axially with respect to the deflation solenoid valve **401**, thus they can only be connected in a bridged manner by an injection-molded core **3** provided with a three-way port. An unnecessary air hole generated by the bridging is blocked by a sealing ball **12** mounted therein by interference fit.
(34) As shown in FIG. **6**, here, the second part region **417***b* of the inflation solenoid valve **402** and the first part region **415***a* of the deflation solenoid valve **401** may be connected by a connecting channel **416** instead of being connected by the three-way port, so that the second part region **417***b* of the inflation solenoid valve **402** communicates directly with the first part region **415***a* of the deflation solenoid valve **401**. In this way, the effect of simplifying the structure can be achieved. In other embodiments, the inflation solenoid valve **402** and the deflation solenoid valve **401** may be connected by other connecting components.
(35) An air inlet nozzle is mounted in the first part region **415***b* of the inflation solenoid valve **402**, and the air inlet nozzle is connected to the side wall of the injection-molded core **403** via a U-shaped iron **11**. An air outlet nozzle is mounted in the third part region **418** of the deflation solenoid valve **401**, and the side wall of the air outlet nozzle is connected to the side wall of the injection-molded core **403** via the U-shaped iron **11**. Here, the air inlet nozzle of the inflation solenoid valve **402** is configured to be connected to the air outlet of the air source, and the air outlet nozzle of the inflation solenoid valve **402** is configured to be connected to the inflated bag. The U-shaped iron **11**

not only connects the air outlet nozzle and the air inlet nozzle to a frame **8** of the composite solenoid valve, but also has the effect of increasing the magnetic field.
(36) As shown in FIG. **5**, the metal valve core **406** of the inflation solenoid valve **402** and/or the deflation solenoid valve **401** is of one of a flat type and a spline type. There is a gap between the flat or spline-type metal valve core **406** and the housing of the solenoid valve. The gap serves as an air path channel between the first part region and the second part region of the solenoid valve.
(37) As shown in FIG. **1**, the cavity of the inflation solenoid valve **402** has a lower end acting as the first part region **415***b*, and an upper end acting as the second part region **417***b*. A silicone cap **410**, a metal valve core **406**, and a spring **409** are arranged in this order from top to bottom in the cavity. When the inflation solenoid valve **402** is powered on, the silicone cap **410** moves down along with the metal valve core **406**. The first part region **415***b* is equipped with an air inlet nozzle. The air inlet nozzle is connected to the side wall of the injection-molded core **403** via the U-shaped iron **11**. The air inlet nozzle is connected to the inflation solenoid valve **402** and then connected to the air outlet of the air source to achieve the inflation function. An airflow flows through the deflation solenoid valve **401** and then into the inflated bag, thereby achieving the inflation function. The cavity of the deflation solenoid valve **401** with the same structure as the inflation solenoid valve **402** has a lower left end acting as the first part region **415***a*, a lower right end acting as the third part region **418**, and an upper end acting as the second part region **417***a*. A silicone cap **410**, a metal valve core **406**, and a spring **409** are arranged in this order from bottom to top in the cavity. The third part region **418** is equipped with an air outlet nozzle. The side wall of the air outlet nozzle is connected to the side wall of the injection-molded core **403** via the U-shaped iron **11**. When the deflation solenoid valve **401** is powered on, the metal valve core **406** of the deflation solenoid valve **401** moves upward, the second part region **417***a* is communicated with the third part region **418**, and the gas in the inflated bag enters from the second part region **417***a* and then is discharged from the third part region **418** to achieve the deflation.
(38) Therefore, in the technical solution of this embodiment, not only the inflation and deflation are achieved by one composite solenoid valve, but also the pressure maintaining function can be achieved when both the inflation solenoid valve **402** and the deflation solenoid valve **401** are closed. Moreover, an upper shell **13** is further provided outside the composite solenoid valve to serve as a protector for the composite solenoid valve. Further, this structure is integrated on a PCB board **14** to further improve the integration of the device and simplify the structure and installation.
(39) The specific working steps are performed as follows:
(40) In this technical solution, a two-position two-way solenoid valve or a two-position three-way solenoid valve is designed as the inflation solenoid valve, and a two-position three-way solenoid valve is designed as the deflation solenoid valve. The two solenoid valves form a solenoid valve module with three working states:
(41) 1. As shown in FIG. **3**, the inflation solenoid valve **402** is powered on, and the deflation solenoid valve **401** is de-energized. The inflation solenoid valve **402** is energized and the metal valve core **406** moves down, so that the gas enters the inflation solenoid valve **402** from the air source through the first part region **415***b* of the inflation solenoid valve **402**, then enters the first part region **415***a* of the deflation solenoid valve **401** from the second part region **417***b* of the inflation solenoid valve **402** through a channel in the injection-molded core **403**, and finally enters the inflated bag from the second part region **417***a* of the deflation solenoid valve **401** to achieve the inflation.
(42) 2. As shown in FIG. **4**, the inflation solenoid valve **402** is de-energized, and the metal valve core **406** of the inflation solenoid valve **402** is pushed up by the spring **409** and moved back to its original position to close the second part region of the inflation solenoid valve **402** and to prohibit the passage of airflow. The deflation solenoid valve **401** is powered on. The deflation solenoid valve **401** is energized such that the metal valve core **406** moves upward. Since no gas passes through the second part region of the inflation solenoid valve **402**, the gas enters the deflation

solenoid valve **401** from the inflated bag and is then discharged from the third part region **418** of the deflation solenoid valve **401** through the second part region **417***a* of the deflation solenoid valve **401** to achieve the deflation.
(43) 3. As shown in FIG. **1**, the inflation solenoid valve **402** is de-energized and the deflation solenoid valve **401** is de-energized. The gas cannot enter the deflation solenoid valve **401**, so that the air pressure in the inflation bag is maintained.
(44) As shown in FIG. **2**, a pump valve module according to this embodiment includes an air pump **4**. Each air outlet of the air pump **4** is equipped with one composite solenoid valve in the foregoing embodiment. The first part region **415***b* of the inflation solenoid valve **402** communicates with the air outlet of the air pump **4**. Optionally, there are four composite solenoid valves, and the four composite solenoid valves are integrated and mounted in a housing formed integrally with the air pump **4**. In other embodiments, any other number of composite solenoid valves may be provided as required.
(45) Optionally, a pressure sensor **5** is mounted in an air path of the pump valve module. The pressure sensor **5** is mounted in the air path of the pump valve module or mounted inside the composite solenoid valve. The pressure sensor **5** monitors the air path system to achieve a more complex control. The pressure sensor **5** may be optionally mounted at a position in any of the following manners.
(46) 1. The pressure sensor **5** is mounted in an air path allowing air supply from the air pump to all the composite solenoid valves and is configured to detect the air pressure of air supplied from the air pump.
(47) 2. The pressure sensors **5** are mounted in air paths between the composite solenoid valves and the inflated bags. In this case, the number of the pressure sensors **5** is identical with the number of the composite solenoid valves, and they are configured to monitor the pressure in the inflated bags, respectively.
(48) 3. The pressure sensors **5** are mounted in both the air path allowing air supply from the air pump to all the composite solenoid valves and in the air paths between the composite solenoid valves and the inflated bags. In this case, the air pressure of each air path in the entire pump valve module can be comprehensively monitored.
(49) This embodiment provides a vehicle seat. The vehicle seat includes a lumbar support and/or side wing supporting structure having an air source supplied by the pump valve module in the foregoing embodiment. The inflatable bag(s) of the lumbar support and/or side wing supporting structure of the vehicle seat is inflated or deflated by a composite solenoid valve. The vehicle seat in this embodiment enables different adjustment and control of multiple inflatable bags.
(50) Reference is made to FIG. **7**, which is a schematic structural diagram of an embodiment of a solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to an embodiment of the present disclosure. A solenoid valve-based pump valve module **20** for providing a massage function for a vehicle seat includes an air distribution layer **23** and further includes a solenoid valve **10** hermetically connected to the air outlet of the air distribution layer **23**. The air distribution layer **23** is connected to the output end of the air pump in the module, communicates with each air outlet of the air pump at its end close to the air pump, and is provided with at least one air outlet at its end far away from the air pump.
(51) It should be noted that the composite solenoid valve according to this embodiment has the same structure as the composite solenoid valve according to the foregoing embodiment.
(52) Reference is made to FIG. **9**, which is a schematic structural diagram of an embodiment of a solenoid valve in an air feeding state according to the present disclosure. This figure shows an energized state of the solenoid valve, and arrows in the figure indicate a gas flow direction.
(53) Reference is made to FIG. **10**, which is a schematic structural diagram of an embodiment of a solenoid valve in an air discharging state according to the present disclosure. This figure shows a non-energized state of the solenoid valve, and arrows in the figure indicate a gas flow direction.

(54) As shown in FIGS. **9** to **11**, the solenoid valve **10** includes a valve body **101**. A solenoid (or electromagnetic) coil **102** is provided around the side wall of the valve body **101**. The valve body **101** is provided with an air distribution chamber **103**, and the two ends of the air distribution chamber **103** act as an air inlet and an air discharge port, respectively. The side wall of the valve body is provided with a branched air passage **104** communicating with the air distribution chamber **103** and facing the same side as the air discharge port. The air discharge port is equipped with an air discharging member **106** in which an air passage **105** is provided, and a valve core **107** built in the air distribution chamber **103** is provided under the air discharging member **106**. A first elastic element **108** is sleeved on both the air discharging member **106** and the valve core **107**. The presence or absence of a gap between the air discharging member **106** and the valve core **107** can be adjusted by switching the solenoid coil **102** to an energized or de-energized state.
(55) Here, the valve body **101** is a basic structure, which may specifically be formed integrally by injection molding. A solenoid coil **102** is provided around the side wall of the valve body. Optionally, reference is made to a schematic structural diagram of an embodiment of a solenoid valve shown in FIG. **11**. An accommodating cavity **110** is provided annularly (or circumferentially) in the outer wall of the valve body **101**, and the solenoid coil **102** is wound in the accommodating cavity **110**. When the solenoid coil is energized, a magnetic field can be generated around the valve body to magnetize components to be magnetized, so that a corresponding action is generated between the magnetized components.
(56) The air distribution chamber is provided inside the valve body and has two ends acting as an air inlet and an air discharge port, respectively. The air inlet is used for being connected to a gas supply structure. The branched air passage **104** is provided in the side wall of the valve body and communicates with the air distribution chamber. An inflatable bag **30** is connected to the free end of the branched air passage.
(57) Based on the above-mentioned design, the air distribution chamber and the branched air passage **104** communicate with each other and can allow the gas to circulate therein. Components to be magnetized, namely, the air discharging member **106** mounted at the air discharge port and the valve core **107** provided in the air distribution chamber, are designed in the air distribution chamber in the valve body. The air discharging member **106** and the valve core are connected by a first elastic element **108**. Optionally, the air discharging member **106** and the valve core **107** are made of iron.
(58) Optionally, a rubber block is embedded in the center of the top of the valve core **107** and is used for sealing the bottom of the air discharging member **106** when the air discharging member **106** attracts the valve core **107**. Optionally, a rubber cap is embedded in the center of the bottom of the valve core **107** and is used for sealing the inlet of the air distribution chamber when the valve core **107** is restored to the original position under the action of the first elastic element **108**.
(59) When the solenoid valve **10** is energized, a magnetic field is established in the valve body **101**. The air discharging member **106** and the valve core **107** act as components to be magnetized. The air discharging member **106** and the valve core **107** are affected by the magnetic field generated by the solenoid coil and have opposite magnetic properties at their proximal ends, so that the fixedly arranged air discharging member **106** can attract the valve core **107**, and there is no gap between the air discharging member **106** and the valve core **107**. At this time, the first elastic element **108** is compressed, and the fixedly arranged air discharging member **106** can attract the valve core **107** so that the position of the valve core **107** is raised, and the air inlet is exposed. An inflation passageway is formed by the air inlet, the air distribution chamber, and the branched air passage **104**. The solenoid valve **10** in this state can inflate the inflatable bag **30** through the inflation passageway.
(60) When the solenoid valve **102** is in a non-energized state, the air discharging member **106** and the valve core **107** are freed from the magnetic field, and the valve core **106** will be moved in a direction away from the air discharging member **106** under the action of the first elastic element

108 until the valve core **107** is returned to the initial position in the air distribution chamber **103** to close the air inlet. In other words, there is a gap between the air discharging member **106** and the valve core **107**, and a deflation passageway is formed by the branched air passage **104**, the air distribution chamber **103**, and the air passage.

(61) In this embodiment, it is possible to adjust whether the air discharging member **106** and the valve core **107** are attracted to each other by adjusting whether the solenoid coil **102** is in an energized state, and finally it is possible to adjust whether there is a gap between the air discharging member **106** and the valve core **107** by switching the solenoid coil **102** to the energized or de-energized state. When the solenoid coil **102** is switched to the energized state, there is no gap between the air discharging member **106** and the valve core **107**, and there is a gas flow in the inflation passageway. When the solenoid coil **102** is switched to the de-energized state, there is a gap between the air discharging member **106** and the valve core **107**, and there is a gas flow in the deflation passageway. The working state of the inflatable bag **30** can be adaptively switched from the inflated state to the deflated state by switching the state of the inflation passageway to the state of the deflation passageway.

(62) The branched air passage **104** is on the same side as the air discharge port, so that the inflation end and the deflation end can be located on the same side. In this way, the overall structural size of the solenoid valve can be significantly reduced, so that the overall structure of the solenoid valve-based pump valve module for providing a massage function for a vehicle seat can be further reduced.

(63) Reference is made to a schematic structural diagram of an embodiment of a solenoid valve-based pump valve module for providing a massage function for a vehicle seat shown in FIG. **13**. In this embodiment, the structure of the air distribution layer is specifically optimized in such a manner that the air distribution layer **23** is provided with at least three air outlets. At least three solenoid valves **10** are provided and are hermetically connected to the air outlets in one-to-one correspondence.

(64) Optionally, there are three air outlets in the air distribution layer, and there are three solenoid valves. Optionally, there are four air outlets in the air distribution layer, and there are four solenoid valves, referring to FIG. **12**. Correspondingly, the number of the solenoid valves corresponds individually to the number of the air outlets of the air distribution layer.

(65) Reference is made to FIG. **15**, which a schematic structural diagram of an embodiment of a solenoid valve in a de-energized state according to the present disclosure. The branched air passage **104** includes an air introducing passage **1041** communicating with the air distribution chamber **103** and an air injection passage **1042** communicating with the side wall of the air passage **1041**. A sealing member **109** is configured in the free end of the air introducing passage **1041**. In the operation of machining the air introducing passage, in order to facilitate machining of a cavity allowing the gas to flow therein, the free end of the air introducing passage is also in a through state so as to facilitate the machining of the internal cavity. When the machining is completed, a sealing member **109** is configured in the free end of the air introducing passage **1041** in order to close the free end of the air introducing passage. Optionally, the sealing member **109** has a spherical structure.

(66) Reference is made to FIG. **16**, which is a schematic structural diagram of an embodiment of a solenoid valve in an energized state according to the present disclosure. A first stop groove **1061** and a second stop groove **1071** are provided annularly in the outer wall of the air discharging member **106** and in the outer wall of the valve core **107**, respectively. The two ends of the first elastic element **108** are locked in the first stop groove **1061** and the second stop groove **1071**, respectively. The first stop groove **1061** and the second stop groove **1071** are designed such that the elastic element can be stably clamped therebetween, and detachment of the elastic element therefrom can be effectively prevented. Optionally, the first elastic element is a spring.

(67) Reference is made to a schematic structural diagram of an embodiment of a solenoid valve

shown in FIG. **17**. The solenoid valve further includes a first U-shaped iron **111** longitudinally straddling the outer wall of the valve body **101**. The first U-shaped iron **111** is provided with through holes allowing the air inlet and the free end of the air discharging member **106** to be exposed.
(68) In this embodiment, the magnetic flux of the magnetic field formed by the solenoid coil can be increased to assist in the above-mentioned operation process. Specifically, the U-shaped iron straddles the side wall of the valve body in FIG. **17**. In other words, it is horizontally arranged and its opening faces the valve body, and its two free ends are fixedly connected to the side wall of the valve body, respectively. The first U-shaped iron **111** is provided with through holes allowing the air inlet and the free end of the air discharging member **106** to be exposed, in order to facilitate the exposure of the air inlet and the air discharge port.
(69) Reference is made to a schematic structural diagram of an embodiment of a solenoid valve shown in FIG. **18**. A sealing groove **1072** is provided at one end of the valve core **107** closer to the air discharging member **106**, and a first sealing ring is provided in the sealing groove **1072**. When the solenoid valve can inflate the inflatable bag via the inflation passageway, the air discharging member **106** and the valve core **107** are affected by the magnetic field generated by the solenoid coil **102** and have opposite magnetic properties at their proximal ends, so that the fixed air discharging member **106** can attract the valve core **107**. A first sealing ring is optionally designed in this embodiment, in order to prevent escape of the gas entering the air passage from the gap between the air discharging member and the valve core.
(70) Reference is made to a schematic structural diagram of an embodiment of a solenoid valve-based pump valve module shown in FIG. **19**. FIG. **20** is a schematic structural diagram of an embodiment of a composite solenoid valve in a pressure maintaining state according to the present disclosure, wherein FIG. **20** shows the specific structure of the composite solenoid valve. At least one air outlet of the air distribution layer **23** is equipped with a composite solenoid valve **40**. The composite solenoid valve **40** includes a deflation solenoid valve **401** and an inflation solenoid valve **402** connected in series and formed integrally by injection molding. A first part region **415***b* of the inflation solenoid valve **402** is connected to an air outlet of an air source. A second part region **417***a* of the deflation solenoid valve **401** is connected to an inflated bag or a bag to be inflated. A second part region **417***b* of the inflation solenoid valve **402** is connected to a first part region **415***a* of the deflation solenoid valve **401**. Inflation is implemented when the inflation solenoid valve **402** is powered on and the deflation solenoid valve **401** is de-energized. The deflation is implemented through a third part region **418** of the deflation solenoid valve **401** when the inflation solenoid valve **402** is de-energized and the deflation solenoid valve **401** is powered on. The composite solenoid valve is placed at the top end of the air source and may constitute one structural body together with the air source. Here, the inflation solenoid valve **402** is a two-position two-way solenoid valve. The second part region **417***b* of the inflation solenoid valve **402** is used for allowing air to be fed into the deflation solenoid valve **401**. The first part region **415***b* of the inflation solenoid valve **402** is used for allowing air to be fed from the air source. The deflation solenoid valve **401** is a two-position three-way solenoid valve. The second part region **417***a* of the deflation solenoid valve **401** is used for inflating the inflated bag. The third part region **418** of the deflation solenoid valve **401** is used for releasing the air. The first part region **415***a* of the solenoid valve **401** is used for receiving the gas from the inflation solenoid valve **402**. The composite solenoid valve is placed at the top end of the air source to form one structural body so as to achieve an integrated compact structural arrangement.
(71) As shown in FIG. **20**, the cavity of the inflation solenoid valve **402** has a lower end acting as the first part region **415***b*, and an upper end acting as the second part region **417***b*. A silicone cap **410**, a metal valve core **406**, and a spring **409** are arranged in this order from top to bottom in the cavity. When the inflation solenoid valve **402** is powered on, the silicone cap **410** moves down along with the metal valve core **406**. The first part region **415***b* is equipped with an air inlet nozzle.

The air inlet nozzle is connected to the side wall of the injection-molded core **403** via a second U-shaped iron **411**. The air inlet nozzle is connected to the inflation solenoid valve **402** and then connected to the air outlet of the air source to achieve the inflation function. An airflow flows through the deflation solenoid valve **401** and then into the inflated bag, thereby achieving the inflation function. Similarly to the structure of the inflation solenoid valve **402**, the cavity of the deflation solenoid valve **401** has a lower left end acting as the first part region **415***a*, a lower right end acting as the third part region **418**, and an upper end acting as the second part region **417***a*. A silicone cap **410**, a metal valve core **406**, and a spring **409** are arranged in this order from bottom to top in the cavity. The third part region **418** is equipped with an air outlet nozzle. The side wall of the air outlet nozzle is connected to the side wall of the injection-molded core **403** via the second U-shaped iron **411**. When the deflation solenoid valve **401** is powered on, the metal valve core **406** of the deflation solenoid valve **401** moves upward, the second part region **417***a* is communicated with the third part region **418**, and the gas in the inflated bag enters from the second part region **417***a* and then is discharged from the third part region **418** to achieve the deflation.
(72) Therefore, in the technical solution of this embodiment, not only the inflation and deflation are achieved by one composite solenoid valve, but also the pressure maintaining function can be achieved when both the inflation solenoid valve **402** and the deflation solenoid valve **401** are closed. Moreover, an upper shell is further provided outside the composite solenoid valve to serve as a protector for the composite solenoid valve. Further, this structure is integrated on a PCB board to further improve the integration of the device and simplify the structure and installation.
(73) In this embodiment, a two-position two-way solenoid valve or a two-position three-way solenoid valve is designed as the inflation solenoid valve, and a two-position three-way solenoid valve is designed as the deflation solenoid valve. The two solenoid valves form a solenoid valve module with three working states:
(74) 1. As shown in FIG. **22**, the inflation solenoid valve **402** is powered on, and the deflation solenoid valve **401** is de-energized. The inflation solenoid valve **402** is energized and the metal valve core **406** moves down, so that the gas enters the inflation solenoid valve **402** from the air source through the connecting channel **416**, then enters the first part region **415***a* of the deflation solenoid valve **401** from the second part region **417***b* of the inflation solenoid valve **402** through a channel in the injection-molded core **403**, and finally enters the inflated bag from the second part region **417***a* of the deflation solenoid valve **401** to achieve the inflation. **2**. As shown in FIG. **21**, the inflation solenoid valve **402** is de-energized, and the deflation solenoid valve **401** is powered on. The deflation solenoid valve **401** is energized such that the metal valve core **406** moves upward. No gas passes through the connecting channel **416**. The gas enters the deflation solenoid valve **401** from the inflated bag and is then discharged from the third part region **418** of the deflation solenoid valve **401** through the second part region **417***a* of the deflation solenoid valve **401** to achieve the deflation.
(75) 3. As shown in FIG. **20**, the inflation solenoid valve **402** is de-energized and the deflation solenoid valve **401** is de-energized. The gas cannot enter the deflation solenoid valve **401**, so that the air pressure in the inflation bag is maintained.
(76) Reference is made to FIG. **23**, which is a schematic structural diagram of an embodiment of a solenoid valve-based pump valve module of the present disclosure, wherein there are four solenoid valves **10**, and there are two composite solenoid valves.
(77) Reference is made to schematic structural diagrams of an embodiment of a solenoid valve-based pump valve module for providing a massage function for a vehicle seat shown in FIGS. **24** and **25**. The air distribution layer **23** is further provided with a pressure regulating protrusion **21**, and a pressure regulating groove **211** is provided in the pressure regulating projection **21**. A pressure regulating hole **212** communicating with the air distribution layer **23** is provided in the bottom of the pressure regulating groove **211**. An overflow (or relief) valve **22** is configured in the pressure regulating projection **21**.

(78) In the structure of this module, a pressure regulating protrusion communicating with the inside of the air distribution layer is provided in the middle of the air distribution layer, and an overflow valve with a pressure stabilizing function is configured in the pressure regulating protrusion, in order to ensure a stable gas pressure in the air distribution layer. When the air pressure in the air distribution layer is greater than a safety threshold, the gas overflows from the overflow valve so that the air pressure is restored to the safety threshold or lower.
(79) In addition, the specific structure of the overflow valve is further disclosed in this embodiment, referring to FIG. **25**. The overflow valve **22** includes: an overflow column **221** with a bottom extending into the pressure regulating hole **212**, a limit ring **222** provided around the outer wall of the overflow column **221**, a second elastic element **223** sleeved on the side wall of the overflow column **221** and located above the limit ring **222**, and a top cover **224** sleeved on the top of the overflow column **221** and provided with a through slot in its center. A second sealing ring **225** is sleeved on the side wall of the overflow column **221** under the limit ring **222**. Here, the overflow column **221** with a bottom extending into the pressure regulating hole for fixing serves as a support to facilitate mounting of other elements.
(80) The limit ring **222** is arranged around the outer wall of the overflow column **221**. The overflow column is divided into a first column section located above the limit ring and a second column section located under the limit ring. The first column section has a side wall around which a second elastic element **223** is wrapped, and a top end wrapped in a top cover **224** provided with a through slot in its inside. A pressure regulating gap is provided between the first column section and the wall of the through slot. The bottom of the through slot is engaged with the top of the second elastic element. A second sealing ring **225** is sleeved on the second column section.
(81) Based on the above-mentioned design, when the air pressure in the air distribution layer does not exceed the threshold, the limit ring on the overflow column is brought into close contact with the second sealing ring under the action of the second elastic element to prevent air release from the overflow valve. When the air pressure in the air distribution layer exceeds the threshold, the airflow enters a first gap between the bottom of the overflow column and the pressure regulating hole and lifts the limit ring so that the first gap communicates with the pressure regulating groove, and then the airflow is discharged to the outside of the air distribution layer through the pressure regulating gap to achieve the purpose of pressure regulation and ensure a stable gas pressure in the air distribution layer.
(82) Furthermore, the solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to this embodiment has the same basic structure as the solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to the foregoing embodiment, therefore the same parts will not be described in detail here.
(83) In addition, reference is made to a schematic structural diagram of an embodiment of a solenoid valve-based pump valve module for providing a massage function for a vehicle seat shown in FIG. **8**. Specifically, in the solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to this embodiment, at least one air outlet of the air distribution layer **23** is equipped with an air path **24** communicating therewith, which has a hollow tubular structure and has an open free end. In the specific structure of the above-mentioned module, an air path connected to at least one air outlet of the air distribution layer is further designed to be reserved for inflating other components. When it is unnecessary to use the air path, its free end may be blocked.
(84) Optionally, the air path has an inverted L-shaped structure.
(85) Alternatively, the air path has a curved structure and has an unrestricted orientation.
(86) Optionally, a sealing cover is provided at the free end of the air path. When it is unnecessary to use the air path, the air path may be blocked by using the sealing cover or blocked by means of gluing or hot melting.
(87) Reference is made to FIG. **13**, which is a schematic structural diagram of an embodiment of a

solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to the present disclosure.
(88) FIG. **13** shows an overall appearance view. The structure of the air distribution layer is specifically optimized in this embodiment in such a manner that the air distribution layer **23** is provided with at least four air outlets. At least three of the air outlets are hermetically connected to the solenoid valves **10** in one-to-one correspondence. One of the air outlets is equipped with an air path **24** communicating therewith, and the free end of the air path **24** is provided with a sealing cover.
(89) Reference is made to a cross-sectional view shown in FIG. **14**, in which the air path is exposed. Specifically, the air distribution layer has four air outlets. Three of the air outlets are hermetically connected to the solenoid valves **10** in one-to-one correspondence. The last air outlet is equipped with an air path **24** communicating therewith, and the free end of the air path **24** are provided with a sealing cover.
(90) The number of the solenoid valves corresponds individually to the number of the inflatable bags. Air can be supplied to different inflatable bags at the same time to achieve a better massage effect.
(91) In addition, reference is made to FIG. **26**, which is a schematic structural diagram of another embodiment of a solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to the present disclosure.
(92) The air distribution layer **23** is further provided with a pressure regulating protrusion **21**, and a pressure regulating groove **211** is provided in the pressure regulating projection **21**. A pressure regulating hole **212** communicating with the air distribution layer **23** is provided in the bottom of the pressure regulating groove **211**. An overflow (or relief) valve **22** is configured in the pressure regulating projection **21**.
(93) In the structure of this module, a pressure regulating protrusion communicating with the inside of the air distribution layer is provided in the middle of the air distribution layer, and an overflow valve with a pressure stabilizing function is configured in the pressure regulating protrusion, in order to ensure a stable gas pressure in the air distribution layer. When the air pressure in the air distribution layer is greater than a safety threshold, the gas overflows from the overflow valve so that the air pressure is restored to the safety threshold or lower.
(94) This embodiment further provides a specific structure of a vehicle seat, which includes the solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to any of the foregoing embodiments.
(95) The solenoid valve-based pump valve module for providing a massage function for a vehicle seat according to this embodiment is applicable to a vehicle seat, so that the vehicle seat has a more compact internal space structure so as to form an optimized internal structural layout.
(96) The above description is merely illustrative of preferred embodiments of the present disclosure and illustrative of the utilized technical principles. It should be understood by those skilled in the art that the scope of the invention involved in the present disclosure is not limited to the technical solutions formed by specific combinations of the foregoing technical features, and should also encompass other technical solutions formed by any combinations of the foregoing technical features or their equivalents without departing from the inventive concept, e.g., technical solutions formed by replacing the foregoing features with the technical features having similar functions disclosed in (but not limited to) the present disclosure or vice versa.
INDUSTRIAL APPLICABILITY
(97) The present disclosure provides a pump valve module and a method for operating the same, a solenoid valve-based pump valve module for providing a massage function for a vehicle seat, and a vehicle seat. The solenoid valve module enables different control of multiple inflatable bags with one air source. Moreover, an air pressure maintaining function is added to the inflatable bags. The

pump valve module has a compact structure, can be easily mounted, has stable functions, and enables the cooperation of multiple units.

Claims

1. An operating method based on a pump valve module, wherein the pump valve module comprises an air pump and a composite solenoid valve, wherein each air outlet of the air pump is equipped with one composite solenoid valve; the composite solenoid valve comprises a deflation solenoid valve and an inflation solenoid valve connected in series and formed integrally by injection molding; a first part region of the inflation solenoid valve is connected to an air outlet of an air source; a second part region of the deflation solenoid valve is connected to an inflated bag; a second part region of the inflation solenoid valve is connected to a first part region of the deflation solenoid valve; inflation is implemented when the inflation solenoid valve is powered on and the deflation solenoid valve is de-energized; and deflation is implemented through a third part region of the deflation solenoid valve when the inflation solenoid valve is de-energized and the deflation solenoid valve is powered on; and the first part region of the inflation solenoid valve communicates with air outlets of the air pump; at least one composite solenoid valve is provided and is mounted integratedly in a housing formed integrally with the air pump; the operating method comprises following working states: a first working condition, wherein the inflation solenoid valve is powered on, the deflation solenoid valve is de-energized, the inflation solenoid valve is energized, and a metal valve core is moved down, so that a gas enters the inflation solenoid valve from the air source through a connecting channel, enters the first part region of the deflation solenoid valve from the second part region of the inflation solenoid valve through a channel in an injection-molded core, and enters the inflated bag from the second part region of the deflation solenoid valve; a second working condition, wherein the inflation solenoid valve is de-energized, the deflation solenoid valve is powered on, the deflation solenoid valve is energized, a metal valve core is moved upward, and the connecting channel is closed, so that the gas enters the deflation solenoid valve from the inflated bag and is discharged from the third part region of the deflation solenoid valve via the second part region of the deflation solenoid valve; and a third working condition, wherein the inflation solenoid valve is de-energized, the deflation solenoid valve is de-energized, and the deflation solenoid valve is closed and configured to maintain air pressure in the inflated bag.
2. The operating method based on a pump valve module according to claim 1, wherein the second part region of the inflation solenoid valve is located at a structural connecting portion in a middle of the composite solenoid valve and is connected, through an injection-molded core, to and staggered axially with respect to the first part region of the deflation solenoid valve; a three-way port is provided in the injection-molded core; the three-way port has an end connected to the second part region of the inflation solenoid valve, an end connected to the first part region of the deflation solenoid valve, and the other end sealed by a sealing ball.
3. The operating method based on a pump valve module according to claim 2, wherein an air outlet nozzle is mounted in the third part region of the deflation solenoid valve, and a side wall of the air outlet nozzle is connected to a side wall of the injection-molded core via a U-shaped iron.
4. The operating method based on a pump valve module according to claim 2, wherein a metal valve core of the inflation solenoid valve and/or the deflation solenoid valve is of one of a flat type and a spline type.
5. The operating method based on a pump valve module according to claim 2, wherein the inflation solenoid valve is a two-position two-way solenoid valve or a two-position three-way solenoid valve, and the deflation solenoid valve is a two-position three-way solenoid valve.
6. The operating method based on a pump valve module according to claim 2, wherein an air inlet nozzle is mounted in the first part region of the inflation solenoid valve, and the air inlet nozzle is

connected to a side wall of the injection-molded core via a U-shaped iron.

7. The operating method based on a pump valve module according to claim 2, further comprising a pressure sensor, wherein the pressure sensor is mounted in an air path of the pump valve module or mounted inside the composite solenoid valve.

8. The operating method based on a pump valve module according to claim 1, wherein the inflation solenoid valve is a two-position two-way solenoid valve or a two-position three-way solenoid valve, and the deflation solenoid valve is a two-position three-way solenoid valve.

9. The operating method based on a pump valve module according to claim 1, wherein an air inlet nozzle is mounted in the first part region of the inflation solenoid valve, and the air inlet nozzle is connected to a side wall of the injection-molded core via a U-shaped iron.

10. The operating method based on a pump valve module according to claim 1, further comprising a pressure sensor, wherein the pressure sensor is mounted in an air path of the pump valve module or mounted inside the composite solenoid valve.